



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SINTHIYA NATESAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

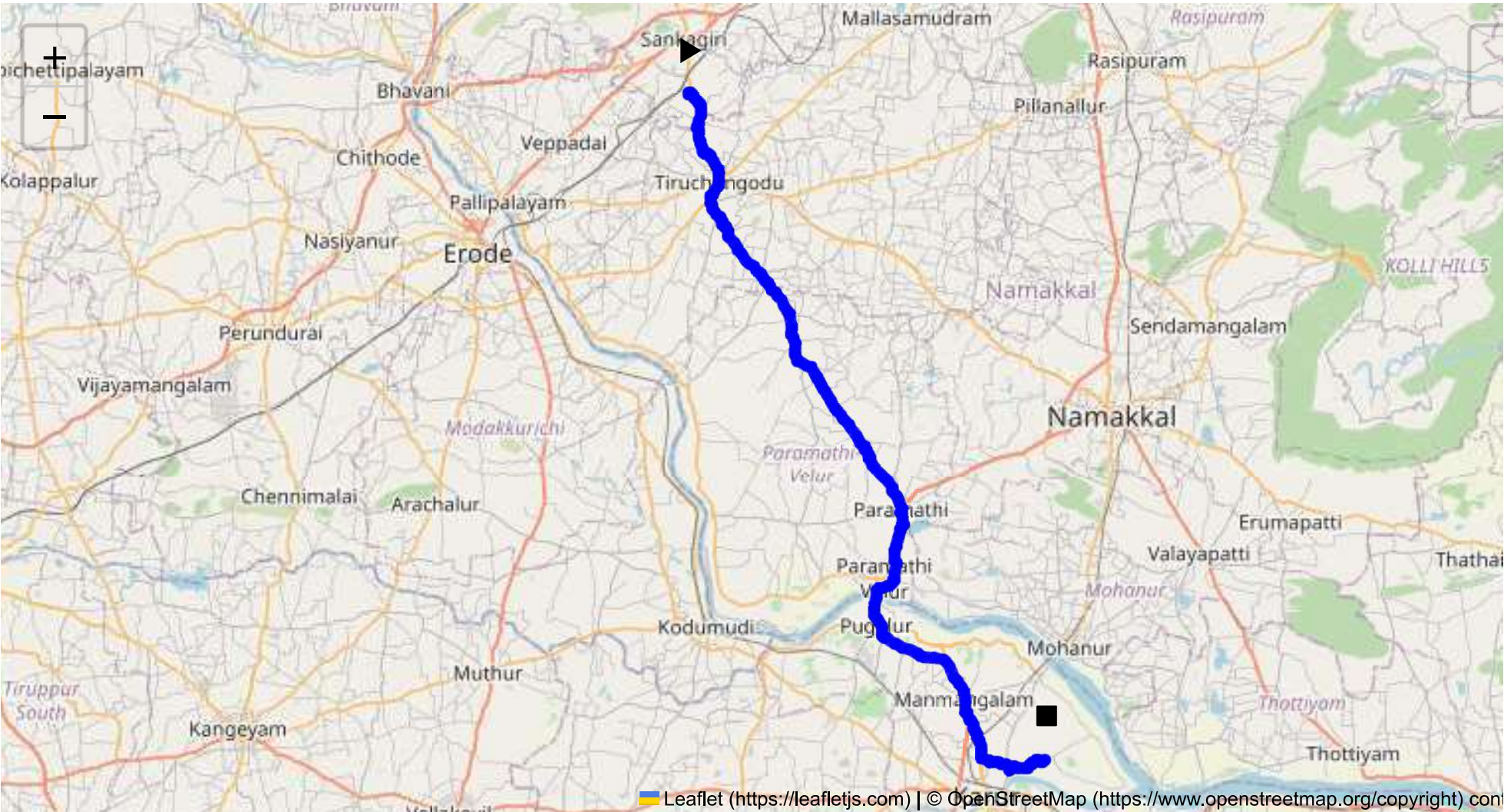
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 67.76 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (10.983946, 78.121374)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The specified route from Sangagiri to Panjamadevi Kaalipalayam covers a distance of approximately 67.76 kilometers and typically takes 1.42 hours under normal conditions. The route traverses through rural and semi-urban areas in Tamil Nadu, primarily using state highways and local roads, notable for intersecting

important towns and agricultural regions. The drive involves navigating various turns and road types, from well-maintained highways to narrower rural paths.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with distinct dry and wet seasons. The northeast monsoon from October to December could lead to heavy rainfall, resulting in waterlogged roads and reduced visibility. The summer months, from March to June, often include intense heat, which can affect driver alertness and vehicle performance. It's crucial to prepare for fog in the early mornings during winter months, which may pose visibility challenges.

3. Analysis of Traffic Patterns

The route might face congestion near towns such as Sankari due to local traffic, especially during peak hours in the morning (7:30 AM - 9:00 AM) and evening (5:30 PM - 7:30 PM). Long stretches of rural roads might witness less traffic but increased farm vehicle activity. Weekends and local festival days can see increased traffic volumes due to regional markets and events.

4. Assessment of Road Quality and Infrastructure

The primary highways are generally in good condition with periodic maintenance, but local roads may present challenges such as potholes, narrow lanes, and inadequate signage. Limited lighting can also be a concern for night travel. It's important to remain cautious at poorly marked intersections and unguarded railway crossings.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, the route offers limited alternative major highways due to infrastructure constraints. However, staying on state highways as much as possible is advisable, with potential detours along NH44 or NH544 if situations demand. It is important to keep updated maps and navigation tools ready to identify other possible detours on less-traveled paths.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is tightly regulated, requiring proper documentation and adherence to national and local guidelines, including specific timings for travel (often restricting movement through highly populated areas during peak hours) and mandatory safety equipment.

7. Overview of Historical Incidents

There are occasional reports of heavy vehicles facing accidents due to overloaded conditions and driver fatigue. However, documented cases specifically involving hazardous materials are rare. Local authorities emphasize safety audits and compliance checks to mitigate such risks.

8. Environmental Considerations and Sensitive Areas

The region includes agricultural lands and areas of natural beauty where hazardous materials might affect the environment if incidents occur. Particular attention should be paid near water bodies and protected green areas to prevent contamination risks.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable along the primary state highways but may become spotty in remote rural areas. Operators should ensure all communication devices are fully charged and backed up by GPS units.

10. Estimated Emergency Response Times

Response times may vary with urban areas seeing quicker reactions, estimated within 20-30 minutes, while rural segments might take 45 minutes to an hour. Proactive communication with local emergency services can improve response readiness.

11. Overall Summary of Risk Assessment

The route poses moderate risks primarily emanating from weather conditions, local traffic congestion, and variable road quality. Maintenance of strict adherence to safety protocols for hazardous materials, coupled with advanced planning for all potential issues, including alternate routing and emergency contacts, is essential for risk management. Regular communication and updates on traffic and weather situations via mobile alerts are advised to ensure timely responses to emerging hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16764, 78.01565	30 KM/Hr
7	Turn	High	11.15573, 78.02176	15 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	10.98525, 78.07657	30 KM/Hr
11	Turn	High	10.98502, 78.07675	15 KM/Hr
12	Turn	Medium	10.98538, 78.07910	30 KM/Hr
13	Turn	High	10.98105, 78.09484	15 KM/Hr
14	Blind Spot	Blind Spot	10.97776, 78.09597	10 KM/Hr
15	Turn	Medium	10.97971, 78.09917	30 KM/Hr
16	Turn	Medium	10.97878, 78.10821	30 KM/Hr
17	Turn	Medium	10.97884, 78.10836	30 KM/Hr
18	Turn	Medium	10.97974, 78.11011	30 KM/Hr
19	Turn	Medium	10.98122, 78.11110	30 KM/Hr
20	Turn	High	10.98164, 78.11315	15 KM/Hr
21	Turn	Medium	10.98250, 78.11346	30 KM/Hr
22	Turn	Medium	10.98457, 78.11549	30 KM/Hr
23	Turn	Medium	10.98465, 78.11595	30 KM/Hr
24	Turn	Medium	10.98449, 78.11650	30 KM/Hr
25	Turn	Medium	10.98369, 78.11700	30 KM/Hr
26	Turn	Medium	10.98332, 78.11883	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	school	Cheran Matriculation Higher Secondary School	10.9971306, 78.0743488	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98525, 78.07657



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.98502, 78.07675



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.98538, 78.07910

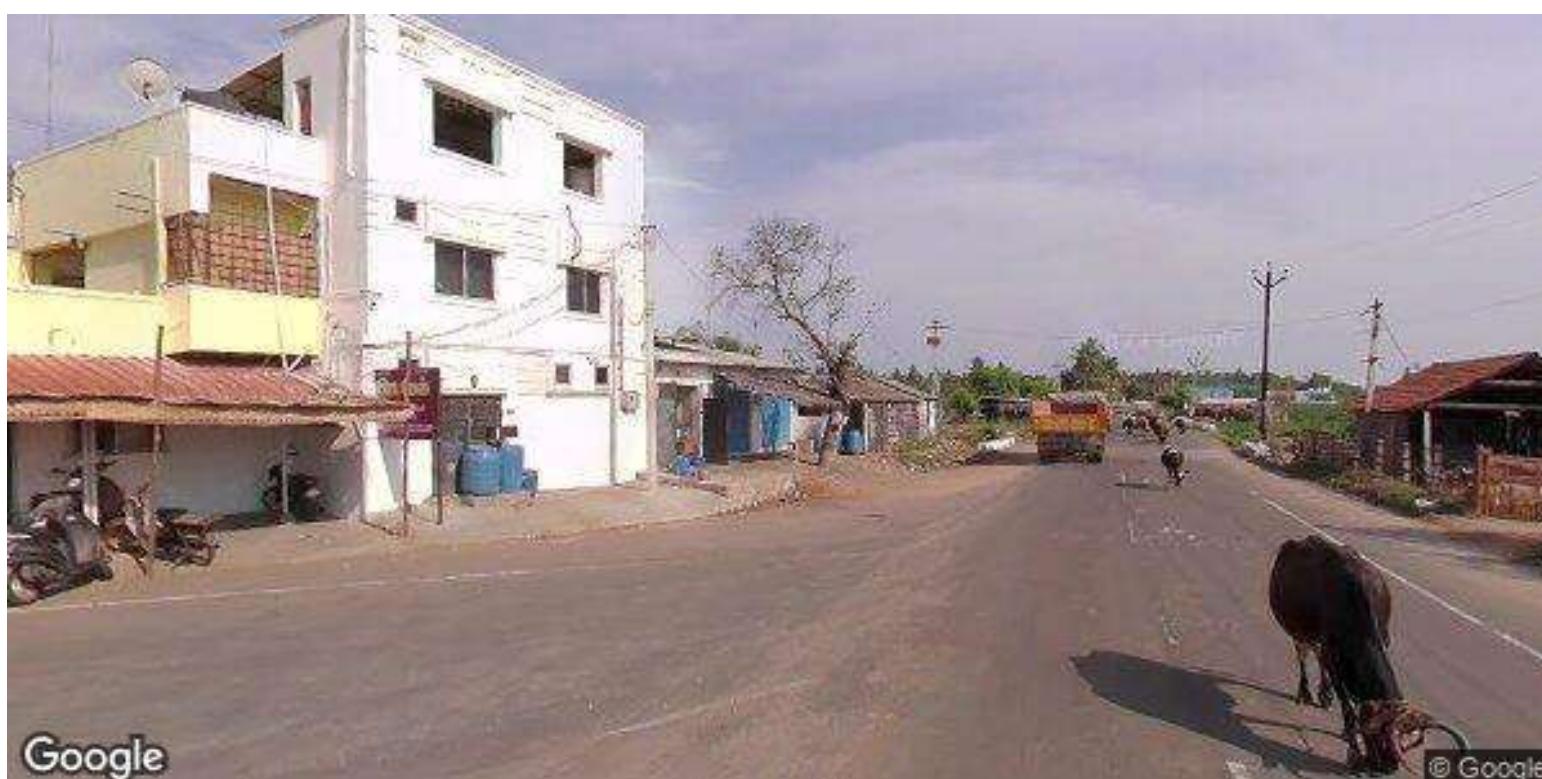


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.98105, 78.09484



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.97776, 78.09597



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.97971, 78.09917



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.97878, 78.10821



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.97884, 78.10836



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.97974, 78.11011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98122, 78.11110



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.98164, 78.11315



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98250, 78.11346



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98457, 78.11549



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98465, 78.11595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98449, 78.11650



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98369, 78.11700



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98332, 78.11883

